

mapping that is converted into a bitmap, which can be saved as a UV texture, in various file formats, such as BMP, TARGA, TIFF, JPEG, SVG, etc.

4. *Painting the texture and applying it:* Once the texture is saved as a marked UV texture map, use your favourite 2D painting application, such as the GIMP, Photoshop, Paintshop, etc., to paint it the way you want to. After that, take the painted texture back to your 3D modeller, in which you can directly apply it to the 3D model. You can test your texture painting by taking a test render from your 3D application and viewing the resulting image. We will look at these steps, in

depth, in the following sections, along with accompanying illustrations to depict how the steps look. We will go into the material assignment first, then the unwrapping, to make our task less cumbersome as we proceed. Note that it is preferable to travel from the top of the plane to the bottom, as you work.

Assigning materials to various parts of the plane

Fire up Wings3D and open the plane model created in last month's article. Once the model is displayed in the viewport, begin by selecting the polygons that form the top of the wings of the plane, as shown in Figure 1. To assign a new 'material' to the selected polygons, click the RMB (right mouse button), and choose the

Material – New option in the *New Material* dialogue. Name it 'wings top', as shown in Figure 2.

You will have the *Material Properties* dialogue come up. Go to the *Diffuse* button and change the colour (see Figure 3). After you are satisfied, click OK. Tap the space bar to de-select the polygons. You will observe that the top polygons have the colour you chose, and a distinct identity, as seen in Figure 4.

Now let's move on to the lower part of the wings. Select the faces that form the lower part of the wings (including the edges of the wings that are left out). Apply a new material to them, giving the material a new colour, and therefore a distinct identity. Be sure you apply a colour that's different

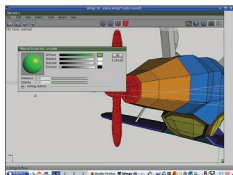


Figure 10: The propeller, after being given an ID

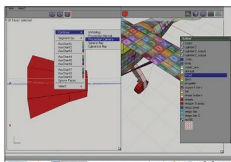


Figure 13: The projection camera applied to the body

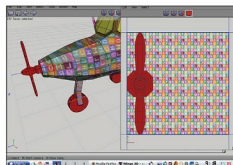


Figure 16: The propeller unwrapped

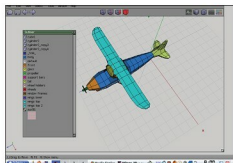


Figure 11: Outliner with materials created

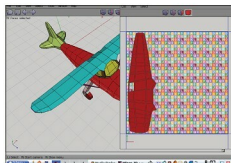


Figure 14: The unwrapped body of plane

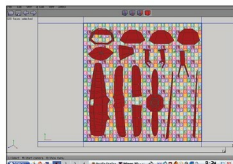


Figure 17: Unwrapped parts assembled together in the central square

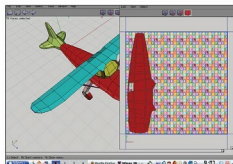


Figure 12: The body in the process of unwrapping

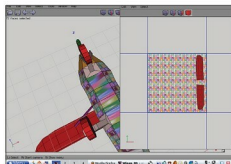


Figure 15: The unwrapped wings placed in position

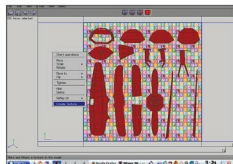


Figure 18: The Create Texture command applied